Physical Design 101

Winter Semiconductor School

Final Project

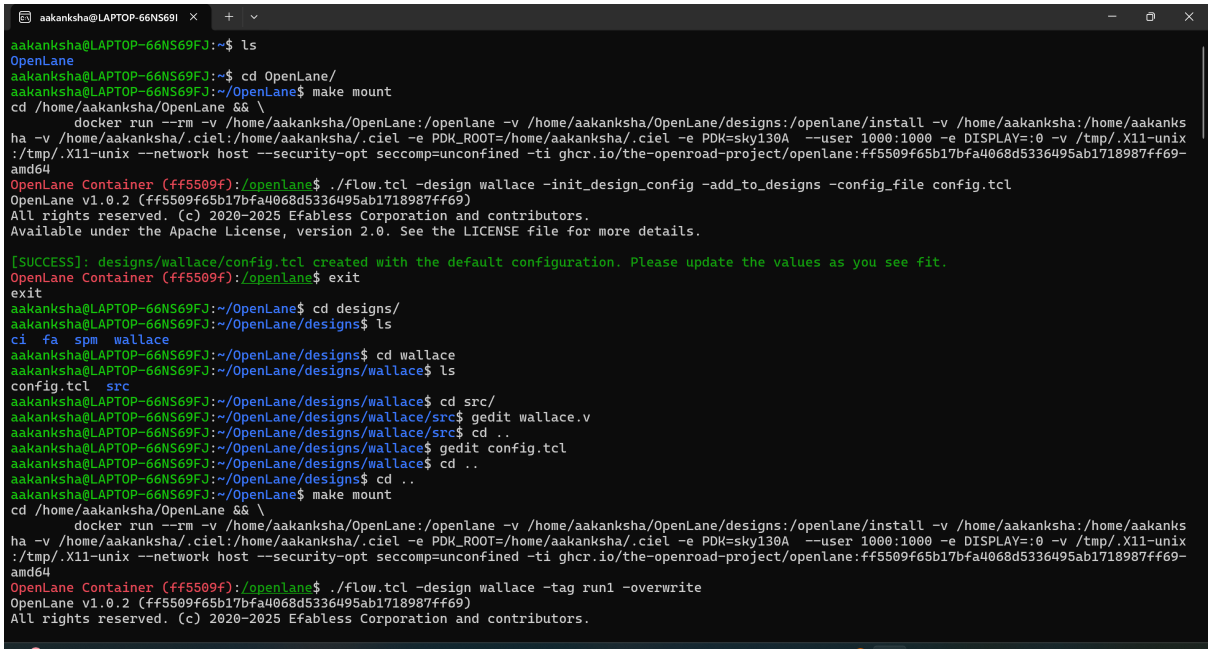
Submitted By: Aakanksha Tambare

Objective

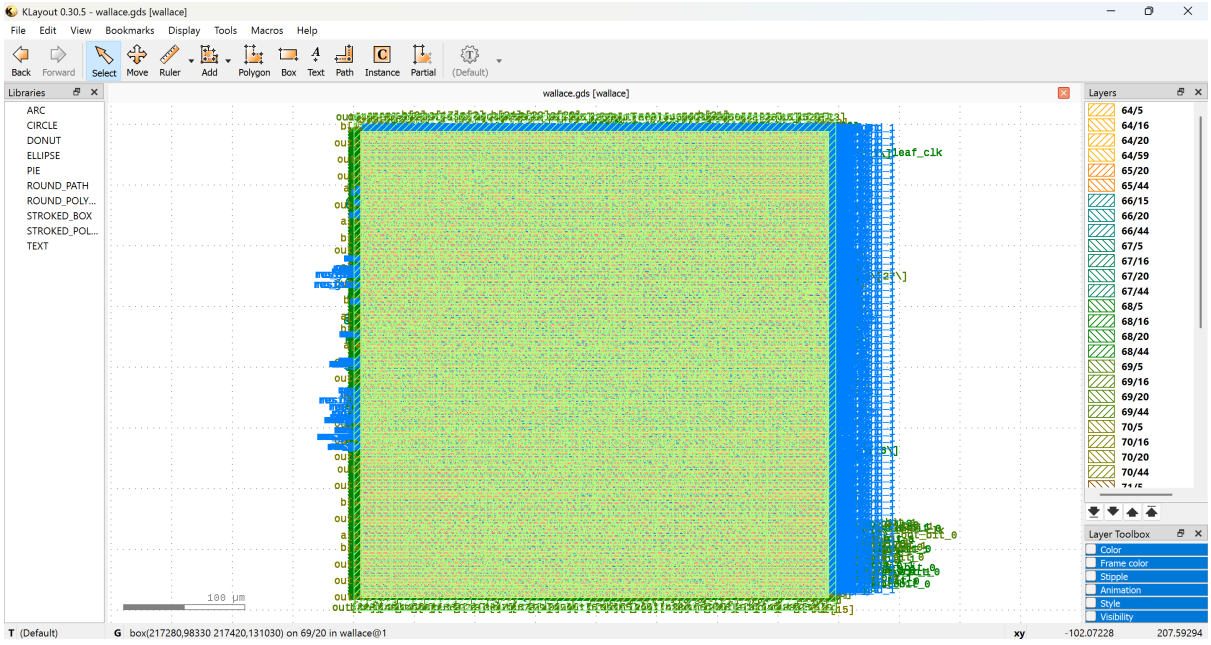
To implement the complete physical design flow of a Wallace Tree Multiplier using OpenLane. The project aims to analyze key physical design metrics such as area, timing, power, and routing quality. Finally, the generated layout is visualized in Blender to gain an intuitive understanding of the chip's physical structure.

OpenLane Flow

- The physical design flow was carried out using the Verilog implementation of the Wallace multiplier as the input design. This Verilog file was provided to OpenLane, which automatically performed the complete RTL-to-GDS flow. All major stages such as synthesis, floorplanning, placement, clock tree synthesis, routing, and final verification were executed successfully without any fatal errors.



- After completing the full OpenLane run, the final GDS file and LEF file were generated. These files represent the physical layout of the Wallace multiplier. The generated GDS and LEF were then opened and analyzed using KLayout, where the overall layout structure, routing layers, and cell placement were visually inspected. This helped in understanding how the logical design is translated into an actual physical chip layout.



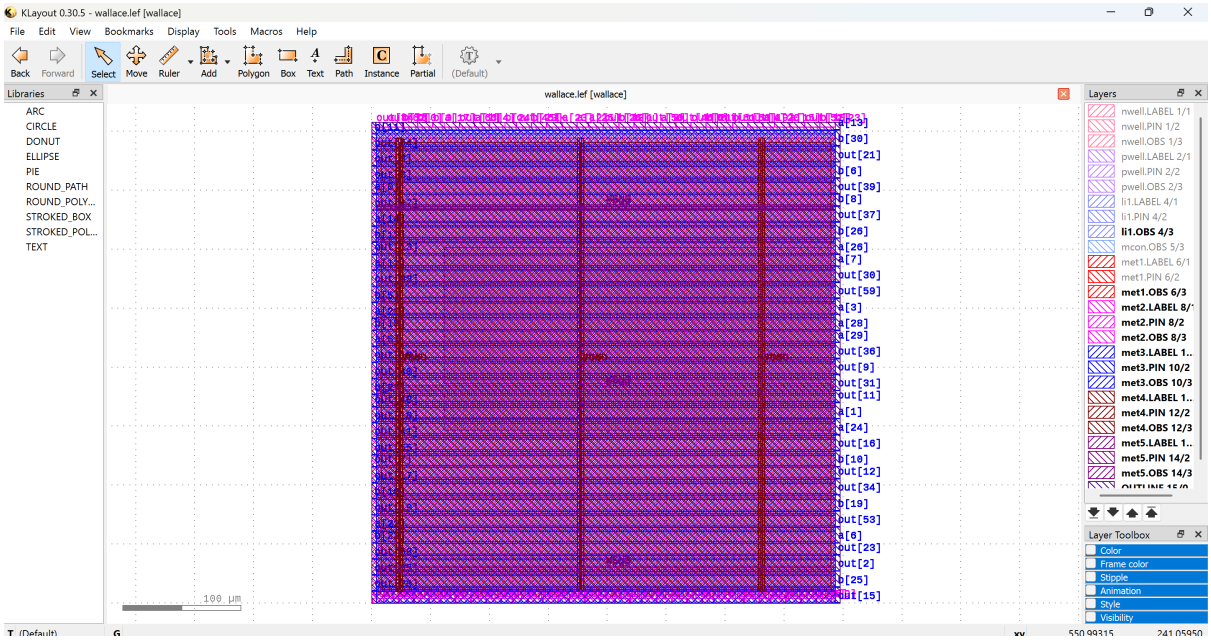


Table 1: Physical Design Metrics for Wallace Tree Multiplier

Metric	Value
DIEAREA (mm^2)	0.162005696625
Core Area (μm^2)	148015.7088
Utilization (%)	
Total Cell Count	20608
WNS (ns)	0
TNS (ns)	0
Critical Path (ns)	5.81
Wire Length	286421
Routing Violations	0
Dynamic Power (μW)	0.01817
Leakage Power (μW)	0.0000000456
Total Runtime	0h9m48s0ms

GDS - STL Conversion

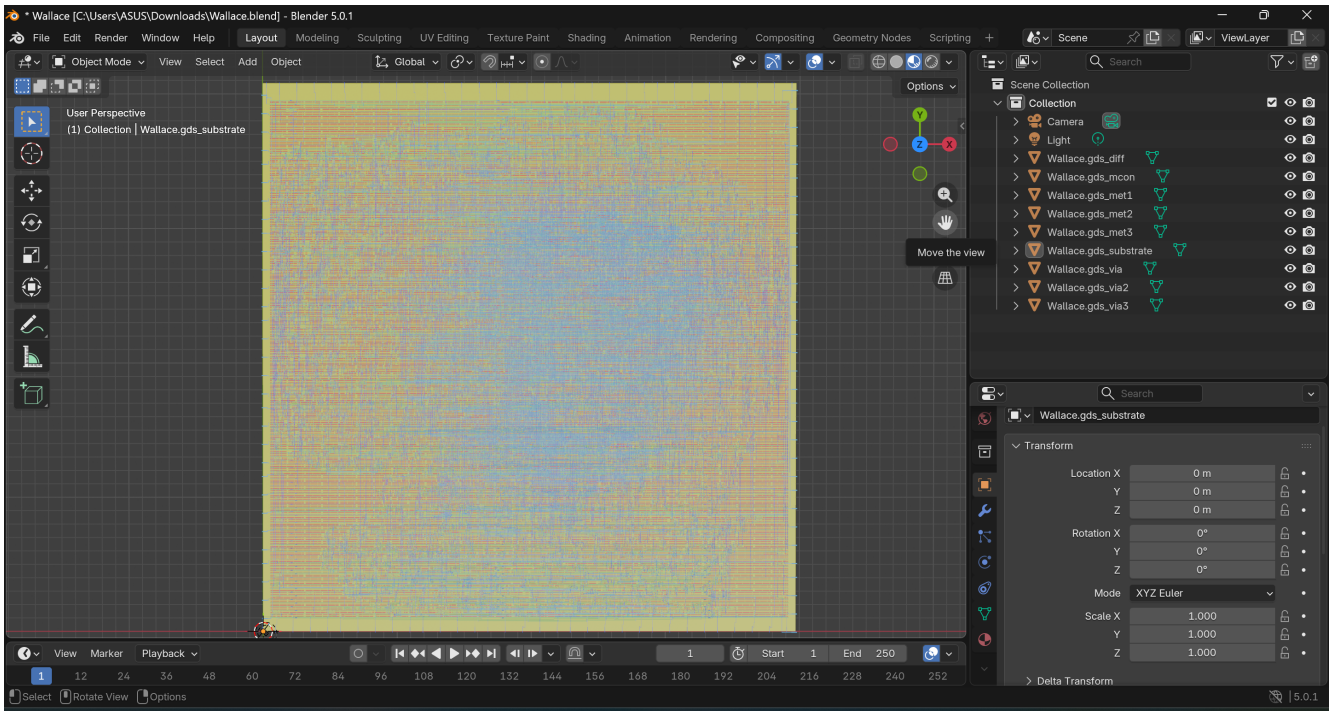
- To transform a GDS (Graphic Data System) layout file into an STL (Stereolithography) format, allowing accurate 3D visualization, modeling, and fabrication using Blender.

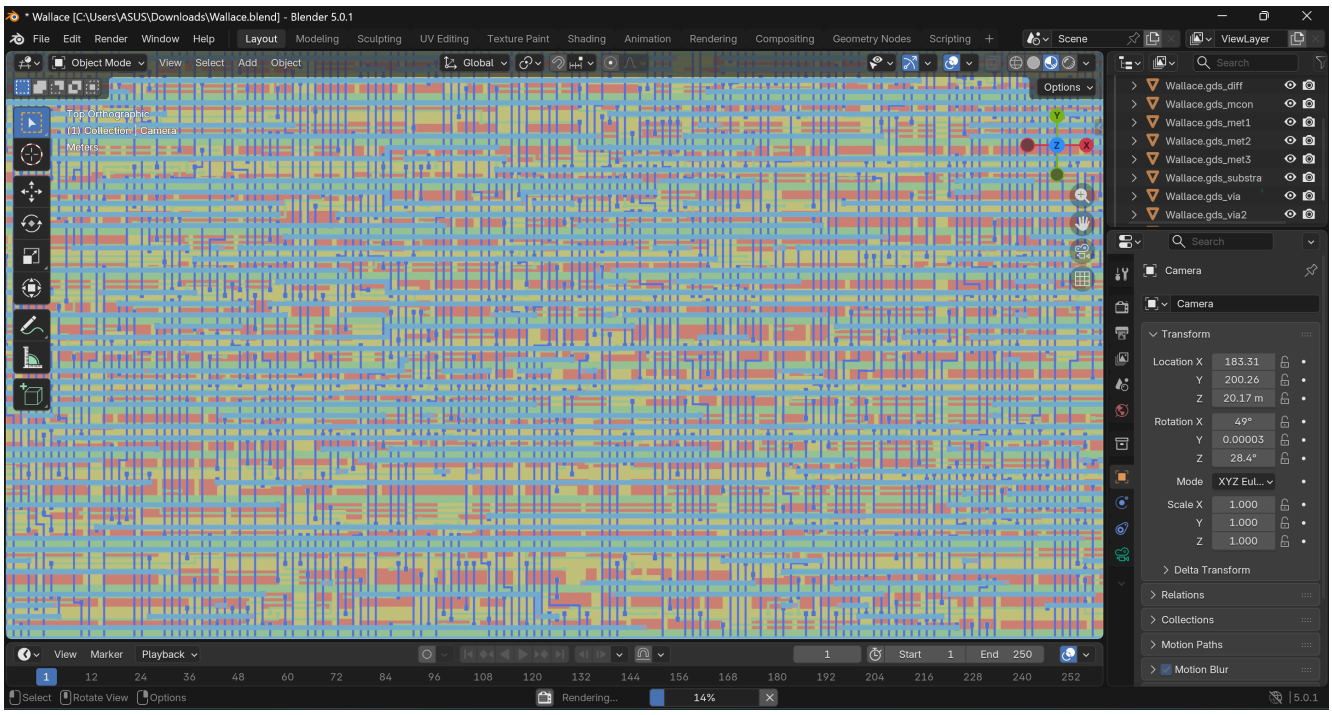
```
aakanksha@LAPTOP-66NS69FJ:~$ cd gdsiistl
aakanksha@LAPTOP-66NS69FJ:~/gdsiistl$ ls
LICENSE                 Wallace.gds_met2.stl        Wallace.gds_via.stl
README.md               Wallace.gds_met3.stl        Wallace.gds_via2.stl
Wallace.gds             Wallace.gds_met4.stl        Wallace.gds_via3.stl
Wallace.gds_diff.stl    Wallace.gds_met5.stl        Wallace.gds_via4.stl
Wallace.gds_li1.stl     Wallace.gds_nwell.stl       example
Wallace.gds_licon.stl   Wallace.gds_poly.stl        gdsiistl.py
Wallace.gds_mcon.stl    Wallace.gds_substrate.stl   requirements.txt
Wallace.gds_met1.stl    Wallace.gds_tap.stl         venv
aakanksha@LAPTOP-66NS69FJ:~/gdsiistl$
```

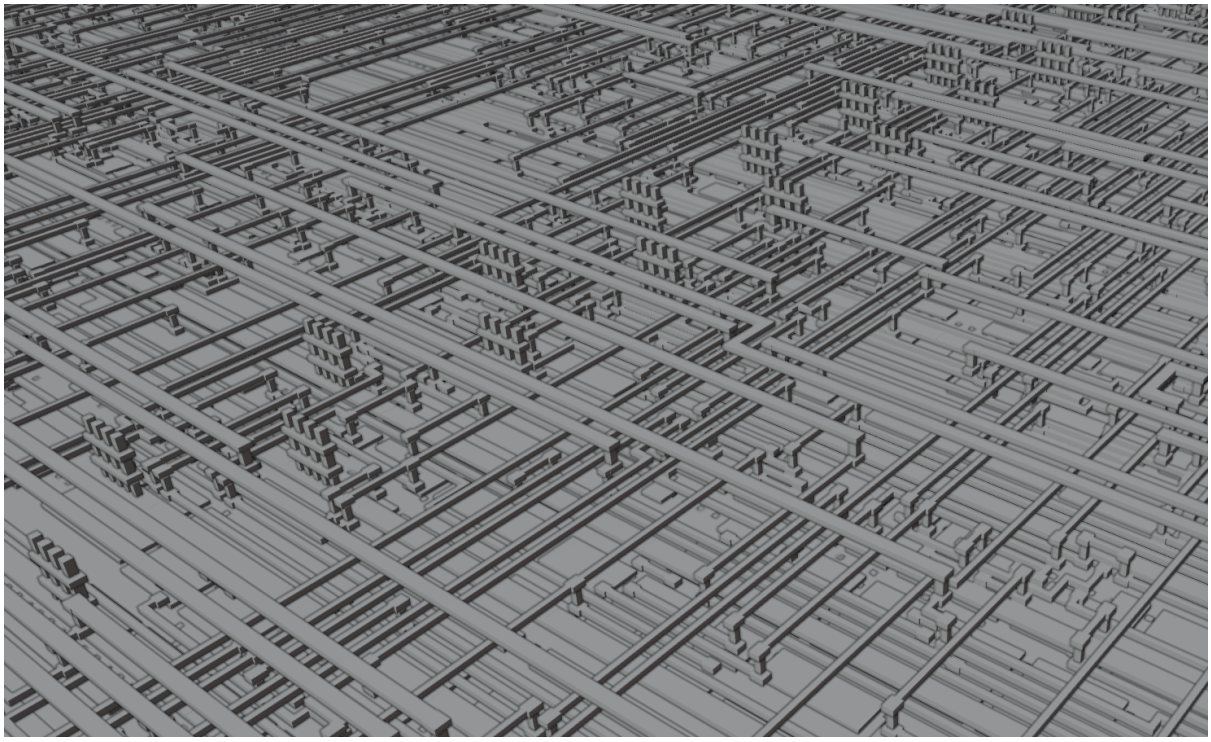
Blender Visualization

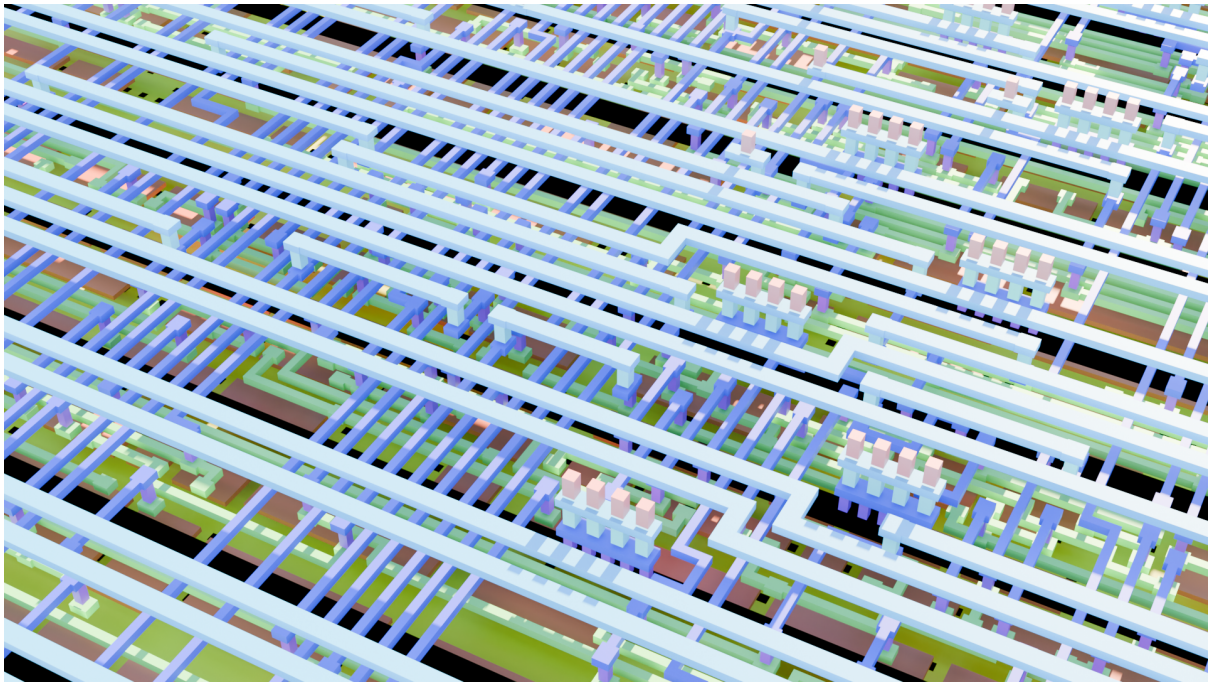
- In this stage of the project, all STL files generated from the GDS layout are imported into Blender to create a structured three-dimensional representation of the design. The substrate is positioned at the lowest level, forming the base of the model, while Metal 1 to Metal 4 layers are arranged above it with increasing Z-offsets to reflect the actual physical stack-up used in fabrication.
- The via layers (Via 1 to Via 3) are placed between their corresponding metal layers to represent vertical electrical interconnections, ensuring correct logical and spatial connectivity within the 3D model. Proper alignment is maintained so that each via accurately connects the intended metal layers.

- Finally, appropriate Z-offsets are assigned to simulate realistic layer spacing, and simple, distinct colors or materials are applied to each layer. This enhances visual clarity, making the structure easier to analyze, model, and present for visualization or fabrication-related purposes.









- Rendered image of Wallace Tree Multiplier

Observation

How routing density changes across metal layers

- When observing routing density, the Wallace Tree Multiplier shows heavier congestion in lower metal layers due to dense partial-product reduction, while higher metal layers are used more for long interconnections and signal spreading. The variation across layers highlights how routing resources are distributed to balance performance and congestion.

Vertical connectivity via vias

- The vertical connectivity through vias clearly shows how signals transition between metal layers, forming stacked connections that enable compact and efficient routing. These vias play a crucial role in maintaining electrical continuity while supporting multi-layer interconnect complexity.

Overall complexity of the multiplier layout

- The overall layout complexity reflects the parallel and hierarchical nature of the Wallace Tree structure, with tightly packed logic blocks and interwoven routing paths. This complexity increases toward the core of the design, demonstrating how high-speed arithmetic architectures demand careful physical organization.

Conclusion

- The complete physical design flow of the Wallace Tree Multiplier was successfully executed using the OpenLane toolchain. After completion of the run, the generated GDS and LEF files were examined using KLayout to verify the layout structure and layer information. The GDS file was then converted into STL format, enabling three-dimensional representation of the design. Subsequently, the STL files were imported layer by layer into Blender, where appropriate Z-offsets were applied to model the vertical stack of layers. Finally, the rendered 3D image was obtained, providing a clear visualization of the multiplier layout and its interconnect structure.